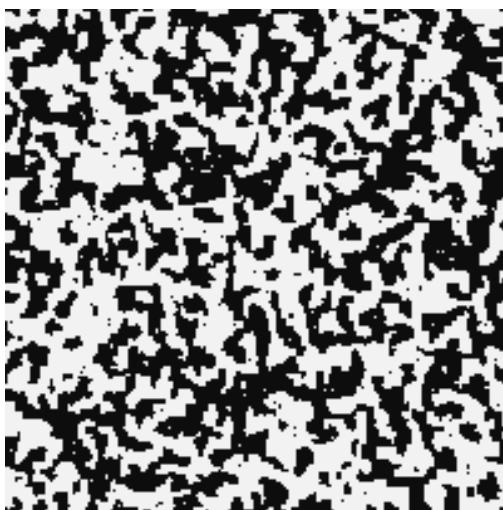


Advanced Statistical Physics

LECTURE NOTES

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Lecture 17: Fluctuation-Response Relations

Given a partition function we can obtain the free energy as $F = -k_B T \ln \mathcal{Z}$ and from there we can find the expectation value of the order parameter,

$$\langle m(\mathbf{r}) \rangle = -\frac{\delta F}{\delta h(\mathbf{r})} = \frac{\int \mathcal{D}m e^{-\beta \mathcal{L}} m(\mathbf{r})}{\int \mathcal{D}m e^{-\beta \mathcal{L}}} \quad (1)$$

The second expression is just the definition of the expectation value. In a similar way, the generalised susceptibility (with spatial dependence) can be found out as,

$$\chi(\mathbf{r} - \mathbf{r}') = \frac{\delta \langle m(\mathbf{r}) \rangle}{\delta h(\mathbf{r}')} = -\frac{\delta^2 F}{\delta h(\mathbf{r}) \delta h(\mathbf{r}')} \quad (2)$$

Using the definition of functional derivatives we can then write the variations,

$$\delta F = - \int d^d r' \langle m(\mathbf{r}') \rangle \delta h(\mathbf{r}') \quad \delta \langle m(\mathbf{r}) \rangle = \int d^d r' \chi(\mathbf{r} - \mathbf{r}') \delta h(\mathbf{r}')$$

We assume that $\mathcal{L}$ has a term $-h(\mathbf{r})m(\mathbf{r})$. Differentiating the expectation value with respect to the field, we have

$$\begin{aligned} \chi(\mathbf{r} - \mathbf{r}') \frac{\delta \langle m(\mathbf{r}) \rangle}{\delta h(\mathbf{r}')} &= \frac{\int \mathcal{D}m e^{-\beta \mathcal{L}} \beta m(\mathbf{r}) m(\mathbf{r}')}{\int \mathcal{D}m e^{-\beta \mathcal{L}}} - \frac{[\int \mathcal{D}m e^{-\beta \mathcal{L}} m(\mathbf{r})] [\int \mathcal{D}m m(\mathbf{r}') \beta e^{-\beta \mathcal{L}}]}{[\int \mathcal{D}m e^{-\beta \mathcal{L}}]^2} \\ &= \frac{1}{k_B T} [\langle m(\mathbf{r}) m(\mathbf{r}') \rangle - \langle m(\mathbf{r}) \rangle \langle m(\mathbf{r}') \rangle] = \beta \mathcal{G}(\mathbf{r} - \mathbf{r}') \end{aligned} \quad (3)$$

where $\mathcal{G}(\mathbf{r} - \mathbf{r}')$ is the correlation function defined in Lec. ???. Hence we get a simple relation,

$$\chi(\mathbf{r} - \mathbf{r}') = \beta \mathcal{G}(\mathbf{r} - \mathbf{r}')$$

Note that we have assumed translational invariance from the beginning, that's why we were writing the functions as a different of the two vectors $\mathbf{r} - \mathbf{r}'$. We can easily use the Fourier transform for translationally invariant systems.

$$\chi(\mathbf{r} - \mathbf{r}') = \int \frac{d^d k}{(2\pi)^d} \tilde{\chi}(k) e^{i\mathbf{k} \cdot (\mathbf{r} - \mathbf{r}')}$$

Using this in the above relation above, we get $\tilde{\chi}(k) = \beta \tilde{\mathcal{G}}(k)$. Now, if the frequency of the external perturbation (like the Fourier decomposition of the magnetic field) tends to zero, χ must reduce to the static isothermal susceptibility,

$$\chi_T = \lim_{k \rightarrow 0} \tilde{\chi}(k) = \beta \tilde{\mathcal{G}}(0) = \beta \int d^d r \mathcal{G}(\mathbf{r})$$

1.1 Calculating the Correlation function (Ornstein-Zernike relation)

Here, we outline a method to calculate the correlation function.

Lecture 18: Some More FLuctuations